

STUDY OF POTASSIUM CITRATE SUPPLEMENTATION IN NORMOCALCIURIC NEPHROLITHIASIS PATIENTS

Muhammad Khalid Khan Niazi, Aziza Khanam*,
Syed Ali Jafer Naqvi and Masood Ahmed Sheikh

*Department of Biochemistry,
University of Karachi
Department of Nephro-urology,
Jinnah Postgraduate Medical Centre, Karachi.

ABSTRACT

The nephrolithiasis patients were treated with aqueous potassium citrate solution orally three times in a day with large quantity of water. After administration of potassium citrate urinary citric acid ($P<0.001$), potassium ($P<0.001$) and pH ($P<0.001$) were significantly increased. Urinary calcium ($P<0.001$), specific gravity ($P<0.001$) and crystalluria were significantly decreased. These observations will be useful in the prevention of calcium nephrolithiasis with hypocitraturia, normocalciuria, as well as controlling calcium nephrolithiasis complicating uric acid lithiasis.

Key words: Normocalciuric, Nephrolithiasis, Citrate.

Introduction

The recurrence rate of renal calculus is very high even if the stones are surgically removed, as approximately 50 percent of patients experience a recurrent stone within five years of the initial episode (Williams 1963). The methods of crushing the stones by using ultrasonic wave or extracorporeal shock wave, which is available more rapidly, are epoch-making treatments, replacing the conventional open surgeries. In both these methods the stones are broken into very small pieces and then extracted or spontaneously eliminated. The Pseudorecurrence from the remaining stone fragments is a problem, and need to be solved. As there is no set therapy which controls the occurrence of calculi on kidney or bladder and its recurrence permanently. Potassium citrate is a new and exciting therapeutic approach, recently introduced for the management of calcium nephrolithiasis. The present study shows the effect of potassium citrate in the normocalciuric nephrolithiasis patients. Jaeger *et al.*, (1986) found hypocitraturia only in normocalciuric idiopathic stone formers, others found reduced urinary excretion of citrate among patients with nephrolithiasis (Rudman *et al.*, 1982, Nicar and Skurla 1983), while reported occurrence of hypocitraturia among various series is

19 to 63 percent (Hodgkinson, 1962, Elliot and Ribeiro, 1972, Menon and Mahle, 1983).

MATERIAL AND METHODS

For the present study 14 women and 33 men were selected who's KuB or IVP, X-ray had shown the presence of upper urinary tract calculus. Those patients with active urinary tract infection, chronic diarrhoeal syndrome, hypokalemia or hyperkalemia were excluded from the study. The endogenous creatinine clearance was more than 70 ml per minute in all patients. Each subject was advised to take ten ml (containing 2.004 g) of potassium citrate solution (containing 102 gm of B.P. grade potassium citrate per 500 ml of distilled water) three times in a day, 30 minutes after each meal for the period of seven days. Pretreatment (control) blood sample and 24 hour urine samples were collected for estimation of biochemical parameters when patients were on random diet and fluid intake without any medication or instruction. After giving potassium citrate supplementation similar type of samples were collected.

ANALYTICAL AND STATISTICAL METHODS

Urinary and blood calcium was analyzed by Biomerieux Kit (Cat. No. 61041), uric acid by phosphotungstate method (Caraway 1955). Citric acid was measured using Natelson technique (Natelson 1961), oxalate was analyzed by Sigma kit (Cat. No. 590-D). Sodium and potassium were measured by flame photometer, inorganic phosphate was estimated using phosphomolybdate method (Delsal and Mannhour, 1982). Lactate dehydrogenase by General Diagnostic kit (LAC-dehydrate, supplied by General Diagnostic Division of Warner-Lambert), alkaline phosphatase activity was measured by Merckotest Alkaline Phosphatase Kit (Cat. No. 344) and gamma-glutamyl transferase activity was estimated by the Boehringer Mannheim Kit (Cat. No. 543080). Creatinine by Jaffe reaction. Urinary protein was measured by sulphosalicylic acid turbidimetric method (Henry *et al.*, 1974) and urinary magnesium by using the Merck Diagnostic Kit (Cat. No. 3338). Serum total protein was estimated by Biuret method, serum albumin was measured by bromocresol green method and urea was measured by Oxime method (Crocker (1967). Significant changes in biochemical values before and after potassium citrate supplementation was assessed by paired 't' test (Walpole 1974).

RESULTS

The patient's age ranged between 11 and 60 years with the mean age of 34.15 ± 1.97 years. On potassium citrate supplementation serum lactate dehydrogenase was significantly increased as compared to pretreatment values.

Serum total protein, albumin, A/G ratio, calcium, inorganic phosphate, alkaline phosphatase, gamma-glutamyl transferase, uric acid and potassium remained unchanged (Table 1).

Table 2 shows the effect of potassium citrate supplementation on urinary constituents. Urinary calcium was significantly decreased ($P<0.001$) after potassium citrate supplementation. Urinary citric acid ($P<0.001$), potassium ($P<0.001$) and pH of urine ($P<0.01$) were significantly increased after the potassium citrate administration. Specific gravity of urine was significantly decreased ($P<0.001$). Urinary inorganic phosphate, urea, creatinine, uric acid, oxalic acid, protein, magnesium, sodium and total urine output remained unchanged after potassium citrate supplementation. Crystalluria was decreased (64%) after potassium citrate supplementation.

DISCUSSION

Many treatments have been used to prevent stone formation. Treatments with orthophosphate (Smith *et al.*, 1988) and magnesium hydroxide (Johnson *et al.*, 1980) have been shown to be effective in preventing the recurrence of calcium stone diseases. These substances are known to reduce the crystalluria by increasing the level of inhibitors of crystallization in urine.

In the present study the effect of potassium citrate supplementation on the blood and urinary constituents were observed in normocalciuric nephrolithiasis patients. Yoshihide and Yasuaki (1986) found that administration of citrate salts induced a slight decrease in both the urinary total and ionized calcium without altering the serum ionized calcium. May and Braun (1985) observed that citrate mixture has no effect on calcium excretion in relapsing calcium oxalate stone-forming patients, when compared with untreated patients. Sakhaee *et al.*, (1983) and Niazi *et al.*, (1993) showed that urinary calcium excretion was decreased during short-term treatment with potassium citrate. The present study showed that the calcium excretion was significantly decreased in the normocalciuric nephrolithiasis patients after potassium citrate supplementation. The potassium excretion was also significantly increased in the urine without producing hyperkalemia. Otto *et al.*, (1982) stated that citrate metabolism may be altered in renal stone disease in general. In normocalciuria, the stone inhibitors Cit: Mg may be deficient. During alkalosis *in vivo*, renal tissue decreases citrate reabsorption and utilization (Anaizi *et al.*, 1986). Preminger *et al.*, (1985) found potassium citrate supplementation raises the urinary citrate, which is considered to be an inhibitor of urinary stone formation. It combines with calcium in urine by chelation and forms a soluble complex salt (Ohkawa and Morimoto 1987). May and Braun (1985) and Yoshihide and Yasuaki (1986) found that administration of citrate salts increased the urinary citrate excretion and concentration. It is the same effect in our study which shows increased urinary citrate excretion in the normocalciuric nephrolithiasis patients after potassium citrate

Table 1 Effect of oral potassium citrate administration on blood constituents of normocalciuric nephrolithiasis patients.

Constituents	Before potassium citrate	After potassium citrate
Total protein gm/l	78.00±0.70 (47)	77.00±0.60 (47)
Albumin (A) gm/l	46.00±0.80 (47)	45.00±0.70 (47)
Globulin (G) gm/l	31.00±0.90 (47)	32.00±0.90 (47)
A/G ratio	16.00±0.60 (47)	15.00±0.80 (47)
Calcium m mol/l	2.35±0.03 (47)	2.30±0.05 (47)
Inorganic phosphate m mol/l	1.17±0.05 (47)	1.24±0.05 (47)
Urea m mol/l	5.23±0.21 (47)	5.05±0.20 (47)
Creatinine m mol/l	88.40±3.54 (47)	79.60±3.54 (47)
Alkaline phosphatase u/l	197.00±7.57 (47)	194.00±8.19 (47)
Gamma-glutamyl transferase u/l	23.70±2.01 (47)	22.80±2.02 (47)
Lactate dehydrogenase u/l	88.70±4.77 (47)	100.80±5.98** (47)
Uric acid m mol/l	0.33±0.05 (47)	0.33±0.01 (47)
Sodium m Eq/l	142.00±0.71 (47)	142.00±0.58 (47)
Potassium m Eq/l	4.30±0.08 (47)	4.30±0.06 (47)

* P<0.02, ** P<0.001.

Table 2 Effect of oral potassium citrate on urinary constituents of normocalciuric nephrolithiasis patients.

Constituents	Before potassium citrate	After potassium citrate
Calcium m mol/d	3.08±0.27 (43)	2.53±0.27*** (43)
Inorganic phosphate m mol/d	270.00±50.00 (43)	241.00±19.60 (43)
Urea nitrogen m mol/d	155.83±14.33 (43)	136.33±10.50 (43)
Creatinine m mol/d	9.83±0.62 (41)	10.64±0.18 (41)
Uric acid m mol/d	8.25±0.50 (46)	9.26±0.67 (46)
Citric acid m mol/d	1.42±0.12 (40)	2.60±0.22*** (40)
Oxalic acid m mol/d	0.34±0.03 (41)	0.33±0.03 (41)
Magnesium m mol/d	4.17±0.26 (45)	4.08±0.39 (45)
Protein gm/d	0.20±0.03 (45)	0.21±0.04 (45)
Sodium m Eq/d	140.00±9.79 (42)	133.00±9.72 (42)
Potassium m Eq/d	26.00±1.49 (41)	48.00±3.18** (41)
pH	5.75±0.09 (47)	6.25±0.09** (47)
Urine volume ml/d	1517.00±133.00 (47)	1690±168.00 (47)
Specific gravity	1.021±0.00126 (47)	1.015±0.0010*** (47)
Crystalluria (No. of patients)	40 (85%)	10 (21%)

*P<0.05, **P<0.01, ***P<0.001.

supplementation. This is due to the decrease in the ionized calcium concentration due to enhanced citrate complex formation with calcium. It is expected that potassium citrate supplementation would be effective in preventing calcium nephrolithiasis with hypocitraturia, normocalciuria, as well as controlling calcium nephrolithiasis. Moreover, this treatment should be effective in the management of uric acid lithiasis because it produces less acidic environment in which uric acid is more soluble and reacts with calcium to form calcium citrate which is more soluble than calcium oxalate in urine. This makes calcium unavailable for precipitation in the urinary tract and thereby reduces stone formation.

Selvam and Rengaraju (1986) had found that the blood lactate dehydrogenase activity is increased in the calcium oxalate stone formers as compared to controls. Lactate dehydrogenase has two role in the endogenous production of oxalate (Sawaki *et al.*, 1966). One is the conversion of glyoxylate to oxalate and second is the conversion of Hydroxypyruvate to L-glycerate, this L-glycerate is later excreted in the urine (Smith 1979). Present study shows that the lactate dehydrogenase activity was significantly increased in the stone formers after potassium citrate supplementation without effecting the oxalic acid excretion suggesting that the increased activity of Lactate dehydrogenase may be involved in the conversion of Hydroxypyruvate to L-glycerate instead of the conversion of glyoxylate to oxalate.

REFERENCES

- Anaizi, N.H., Tullius, J.C., Altamese, J.B. and Steven, J.W., (1986). Renal tissue citrate: Independence from citrate utilization, reabsorption and pH, *Am. J. Physiol.*, 251 (3 part 3) : F547.
- Caraway, W.T., (1955). Determination of uric acid in serum by carbonate method, *Am. J. Clin. Path.*, 25 : 840.
- Crocker, C.L., (1967). Rapid determination of urea nitrogen in serum or plasma without deproteinization, *Am. J. Med. Technol.*, 33 : 361.
- Delsal, J.L. and Mannhour, H., (1982). *Bulletin de la Societe de chimie Biologique*, 40 : 1623. Cited by Wootton, I.D.P. and Freeman, H. *Micro-analysis in Medical Biochemistry*, p. 59.
- Elliot, J.S. and Ribeiro, M.E., (1972). The urinary excretion of citric, hippuric, and lactic acid in normal adults and in patients with calcium oxalate urinary calculus disease, *Invest. Urol.*, 10 : 102.
- Henry, R.J., Cannon, D.C. and Winkelman, J.W., (1974). Chemistry Principles and Technics, *Bioscience Laboratories*, 2nd Edition, p. 543.
- Hodgkinson, A., (1962). Citric acid excretion in normal adults in patients with renal calculus, *Clin. Sci.*, 23 : 203-12.

- Jaeger, P., Portman, L., Jacquet, A.F. and Burchardt, (1986). Indications for urinary citrate determination in idiopathic stone former, *Schweiz. Med. Wochenscher.*, 116(2) : 371.
- Johanson, G., Backman, U., Danielson, B.G., Fellstrom, B., Ljunghall, S. and Wikstorm, B., (1980). Biochemical and clinical effects of the prophylactic treatment of renal calcium stones with magnesium hydroxide, *J. Urol.*, 124 : 770-74.
- May, P. and Braun, J., (1985). Citrate excretion by relapsing calcium oxalate stone formers before and during alkalinizing treatment, *Fortscher. Urol. Nephrol.*, 23 : 386-91.
- Menon, M. and Mahle, C.J., (1983). Urinary citrate excretion in patients with renal calculi, *J. Urol.*, 129 : 1158-60.
- Natelson, S. (1961). Microtechniques of clinical chemistry, 2nd Edition, p. 187.
- Niazi, M.K.K., Khanam, A., Naqvi, S.A.J. and Sheikh, M.A., (1993). Study of potassium citrate treatment of crystalluric nephrolithiasis, *Biomed. and Pharmacother.*, 47 : 25-28.
- Nicar, M. and Skurla, C., (1983). Low urinary citrate excretion in nephrolithiasis, *Urology*, 21(1) : 8-14.
- Ohkawa, T. and Morimoto, S., (1987). Methods for prevention the recurrence of urinary stone disease, *Asian Med. J.*, 30(3) : 143-48.
- Otto, S.P., Scholz, D., Schwille, K., Leutsch, R., Goldberg, I. and Sigel, A., (1982). Citrate in urine and serum and associated variables in subgroups of urolithiasis: Results from an outpatient stone clinic, *Nephron*, 31(3) : 194-202.
- Preminger, G.M., Sakhaee, K., Skurla, C. and Pak, C.Y.C., (1985). Prevention of recurrent calcium stone formation with potassium citrate therapy in patients with distal renal tubular acidosis, *J. Urol.*, 134 : 20-23.
- Rudman, D., Kutner, M.H., Redd, S.C., Waters, W.C., Gerson, G.G. and Bleier, J., (1982). Hypocitraturia in calcium nephrolithiasis, *J. Clin. Endocr. Metab.*, 55 : 1052-57.
- Sakhaee, K., Nicar, M., Hill, K. and Pak, C.Y.C., (1983). Contrasting effects of potassium citrate and sodium citrate therapies on urinary chemistries and crystallization of stone forming salts, *Kidney Int.*, 24(3) : 348-52.
- Sawaki, S., Hattori, N. and Yamada, K. (1966). Reduction of nicotinamide adenine dinucleotide by glyoxylate in animal organs, *J. Vitaminology*, 12 : 303.
- Selvam, R. and Rengaraju, M. (1986). Alterations in red blood cell lactate dehydrogenase isoenzyme pattern in calcium oxalate stone formers, *Indian J. Biochem. Biophys.*, 23(5) : 229-301.
- Smith, L.H., (1979). Enteric hyperoxaluria and other hyperoxaluric states. In: *Urolithiasis*, ed. Coe, F.L., Brenner, B.M. and Stein, J.H. pp. 137-143.

- Smith L.H., Werness, P.G. Van den Berg, C.J. and Wilson, D.M., (1980). Orthophosphate treatment in urolithiasis, *Scand. J. Urol. and Nephrol. Supplement*, 53 : 253-59.
- Walpole, R.E., (1974). *Introduction to Statistics*, 2nd Edition, Macmillan Publishing Co., Inc. Oage, p. 204.
- Williams, R.C., (1963). Long term survey to 538 patients with upper urinary tract stones, *Br. J. Urol.*, 35 : 416.
- Yoshihide, P. and Yasuaki, U., (1986). Impact of acute oral single administration of CG-120 to healthy humans, with special reference to stone-forming substances, *Yakuri to Chiryō*, 14(8) : 5251-72.